

Dr. Utkarsh Jadli

Electronics & Field Engineer

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PROFESSIONAL SUMMARY

Electronics Engineer with a PhD in Power Semiconductor Devices and 4+ years delivering certified hardware systems for safety-critical, harsh-environment applications. Experienced across the full hardware lifecycle — from requirements definition and architecture through PCB design, embedded firmware, verification and validation, and field deployment. Strong background in precision analog front-end design, mixed-signal systems, sensor integration, and environmental test campaigns. Certified hardware designed under IEC 60079 and IEC 80079 standards, with formal DFMEA, requirements traceability, and independent certification authority review. Applying the same engineering rigour — margin analysis, fault tolerance, and first-time-correct design — to electronic systems.

Portfolio: [utkarshjadli-bit.github.io](https://github.com/utkarshjadli-bit)

CORE COMPETENCIES

Verification & Validation (V&V) — Unit, Subsystem & System Level
Safety-Critical Hardware Design — IEC 60079 / IEC 80079 Certified Systems
Mixed-Signal Circuit Design — Analog Front-End Architecture — Embedded Systems
Requirements Definition — DFMEA — Requirements Traceability
Environmental Testing — Thermal, Mechanical & Integration Testing
PCB Design (Altium) — EMI Mitigation — DFM/DFT
Embedded C (STM32, ESP32) — I2C — SPI — UART — Industrial Telemetry
Root Cause Investigation — Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Sigra Pty Ltd — [Brisbane, Australia](#)

Nov 2021 – Feb 2026

Electronics Engineer — Safety-Critical Instrumentation

Verification, Validation & Certification

- Planned and executed full V&V campaigns for a 24-bit strain acquisition system: beam-bending validation against analytical models, 67-hour thermal drift characterisation across a 39.6°C temperature swing, and mechanical testing using UCS equipment.
- Resolved 11 technical queries from an independent certification authority during certification of an intrinsically safe battery management system — covering fault energy calculations, component derating, and worst-case failure scenarios.
- Developed test procedures, formal acceptance criteria, and requirements traceability documentation to close certification requirements under IEC 60079 and IEC 80079.
- Performed DFMEA, worst-case capacitance analysis, and component derating analysis as part of safety case development for hazardous area hardware.

Hardware Design & Systems Engineering

- Architected a 24-bit strain acquisition platform achieving 17.8-bit ENOB and 1.83 $\mu\epsilon$ RMS noise within a constrained 27 mm cylindrical form factor.
- Developed IMU-based borehole survey system achieving $\pm 0.33^\circ$ azimuth, $\pm 0.11^\circ$ inclination, and $\pm 0.25^\circ$ tool-face accuracy; validated HART telemetry over 1.4 km armoured drilling cable across three independent tool builds.
- Designed low-noise analog front-end circuitry including instrumentation amplification, filtering, ADC interfacing, and sensor excitation for high-accuracy measurement under harsh field conditions.
- Designed and debugged multi-layer PCBs in Altium Designer with emphasis on mixed-signal layout, grounding strategy, and EMI mitigation.
- Implemented closed-loop hydrostatic pressure test rig (0–22 MPa) using PID-controlled electropneumatic regulator; characterised aluminium Young's modulus at 70.8 GPa (2.65% error).

Embedded Firmware & Integration

- Developed embedded firmware in C for STM32 and ESP32 microcontrollers: sensor interfacing (I2C, SPI, UART), data acquisition, calibration routines, and HART/RS-485/Modbus telemetry.

- Integrated multi-sensor subsystems (strain gauges, IMUs, thermistors, Hall-effect, ToF) from schematic through field deployment.
- Delivered a \$250k+ ACARP engineering project on schedule during global semiconductor shortages, managing component qualification, COTS evaluation, and adaptive PCB redesign.
- Conducted on-site root cause investigation and troubleshooting across active mining sites under operational time constraints.

RESEARCH & ENGINEERING EXPERIENCE

Griffith University – [Brisbane, Australia](#)

2018 – 2021

PhD Researcher — Power Semiconductor Device Modelling & Characterisation

- Characterised Si MOSFETs, SiC MOSFETs, and GaN HEMTs via I-V and C-V measurements across temperature — device families increasingly adopted in space power conversion and RF systems for their radiation tolerance, high-temperature operation, and switching efficiency advantages over silicon.
- Developed SPICE-based device models including parasitic capacitance extraction and double-pulse test validation for dynamic switching behaviour.

QMNC, Griffith University – [Brisbane, Australia](#)

2019 – 2021

Research Assistant / Fellow

- Designed and debugged PCBs in Altium for laboratory sensor systems and automated measurement setups.
- Supported interdisciplinary teams with electronics integration, hardware debugging, and test workflow improvements.

EDUCATION

Ph.D. — Power Semiconductor Devices, Griffith University (Brisbane)

2018 – 2021

M.Tech — Control Engineering, Graphic Era University (Dehradun)

2015 – 2017

B.Tech — Electrical & Electronic Engineering, Graphic Era University (Dehradun)

2011 – 2015

TECHNICAL SKILLS

Hardware Design: Mixed-Signal Circuit Design, Precision Analog Front Ends, Sensor Excitation & Conditioning, EMI Mitigation, Multi-layer PCB Design (Altium), DFM/DFT

Embedded Systems: STM32, ESP32, Embedded C, Sensor Interfacing (I2C, SPI, UART), Data Acquisition, Calibration Workflows

V&V & Systems Engineering: Requirements Definition, DFMEA, Requirements Traceability, Environmental Testing, Thermal Analysis, Integration Testing, Root Cause Investigation; familiar with DO-254/DO-178 frameworks

Communication Protocols: UART, RS-485, Modbus RTU, HART

Simulation & Modelling: LTspice/SPICE, MATLAB/Simulink, Python (NumPy, SciPy)

PUBLICATIONS & PROFESSIONAL MEMBERSHIP

Published research on power semiconductor device modelling and characterisation (SiC/GaN) and power electronics simulation.

Google Scholar: scholar.google.com.au/citations?user=kmwQvPQAAAAJ

Member: IEEE Power Electronics Society | IEEE Electron Device Society